

EXTENSION 1

Stage 6

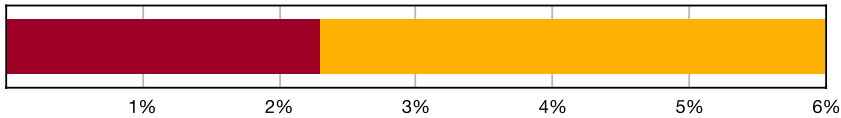
Functions (Ext1), F2 Polynomials (Y11)

Sum, Products and Multiplicity of Roots (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 118.5 minutes (based on allocation of 1.5 minutes per mark)

F2 Polynomials



- Remainder and Factor Theorems
- Sum, Product and Multiplicity of Roots

*SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.

HISTORICAL CONTRIBUTION

- Polynomials* has contributed a solid 6.0%, on average, to new syllabus Ext1 exams since being introduced in 2020.
- This topic has been split into two categories for analysis purposes which are: *1-Remainder and Factor Theorems (2.3%)* and *2-Sum, Products and Multiplicity of Roots (3.7%)*.
- This analysis looks at *Sum, Products and Multiplicity of Roots*.

HSC ANALYSIS - What to expect and common pitfalls

- Sum, Products and Multiplicity of Roots (3.7%)* has surprised to the upside in its contributions to Ext1 new syllabus exams, with longer answer questions appearing each year in the period 2020-23 as well as a multiple choice in 2020 and 2024.
- The removal of *Polynomials* from the new Ext2 syllabus has resulted in *Sum, Products and Multiplicity of Roots* being examined to a greater degree of difficulty than in the past.
- 2023 Ext1 14b*, *2022 Ext1 13d* and *2021 Ext1 11h* provide case in point with low mean marks produced by each question. All new syllabus questions in this area represent important revision.
- Numerous Ext2 past questions that are now examinable within the new Ext1 course are included in the database - a number focus on the multiplicity of roots which we regard as an important revision area.

Questions

1. Functions, EXT1' F2 2015 HSC 4 MC

The polynomial $x^3 + x^2 - 5x + 3$ has a double root at $x = \alpha$.

What is the value of α ?

- A. $-\frac{5}{3}$
- B. -1
- C. 1
- D. $\frac{5}{3}$

2. Functions, EXT1 F2 2024 HSC 1 MC

The polynomial $x^3 + 2x^2 - 5x - 6$ has zeros $-1, -3$ and α .

What is the value of α ?

- A. -2
- B. 2
- C. 3
- D. 6

3. Functions, EXT1 F2 2012 HSC 3 MC

A polynomial equation has roots α, β, γ where

$\alpha + \beta + \gamma = -2, \quad \alpha\beta + \alpha\gamma + \beta\gamma = 3, \quad \alpha\beta\gamma = 1.$

Which polynomial equation has the roots α, β , and γ ?

- A. $x^3 + 2x^2 + 3x + 1 = 0$
- B. $x^3 + 2x^2 + 3x - 1 = 0$
- C. $x^3 - 2x^2 + 3x + 1 = 0$
- D. $x^3 - 2x^2 + 3x - 1 = 0$

4. Functions, EXT1' F2 2012 HSC 5 MC

The equation $2x^3 - 3x^2 - 5x - 1 = 0$ has roots α, β and γ .

What is the value of $\frac{1}{\alpha^3\beta^3\gamma^3}$?

- A. $\frac{1}{8}$
- B. $-\frac{1}{8}$
- C. 8
- D. -8

5. Functions, EXT1' F2 2017 HSC 5 MC

The polynomial $p(x) = x^3 - 2x + 2$ has roots α, β and γ .

What is the value of $\alpha^3 + \beta^3 + \gamma^3$?

- A. -10
- B. -6
- C. -2
- D. 0

6. Functions, EXT1' F2 2018 HSC 3 MC

The cubic equation $x^3 + 2x^2 + 5x - 1 = 0$ has roots, α, β and γ .

Which cubic equation has roots $\frac{-1}{\alpha}, \frac{-1}{\beta}, \frac{-1}{\gamma}$?

- A. $x^3 - 5x^2 - 2x + 1 = 0$
- B. $x^3 - 5x^2 - 2x - 1 = 0$
- C. $x^3 + 5x^2 + 2x + 1 = 0$
- D. $x^3 + 5x^2 - 2x + 1 = 0$

7. Functions, EXT1' F2 2019 HSC 4 MC

The polynomial $2x^3 + bx^2 + cx + d$ has roots 1 and -3 , with one of them being a double root.

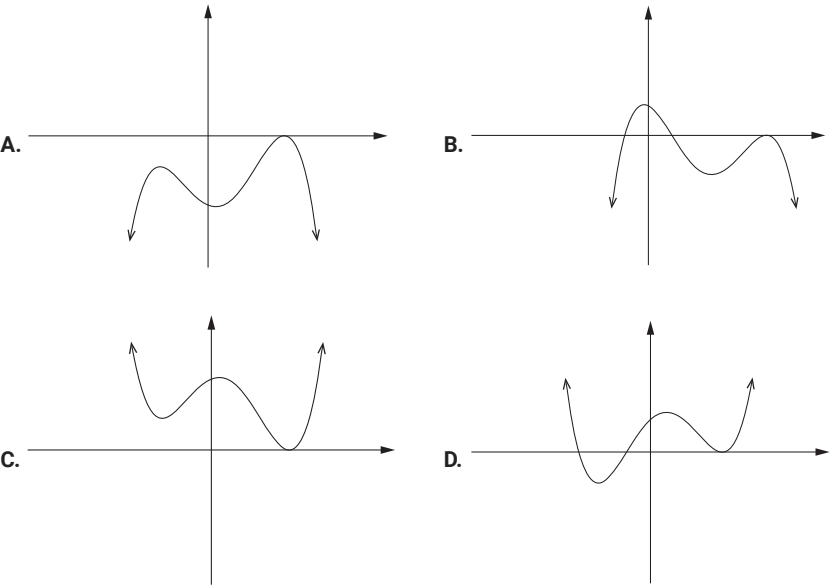
What is a possible value of b ?

- A. -10
- B. -5
- C. 5
- D. 10

8. Functions, EXT1 F2 2020 HSC 5 MC

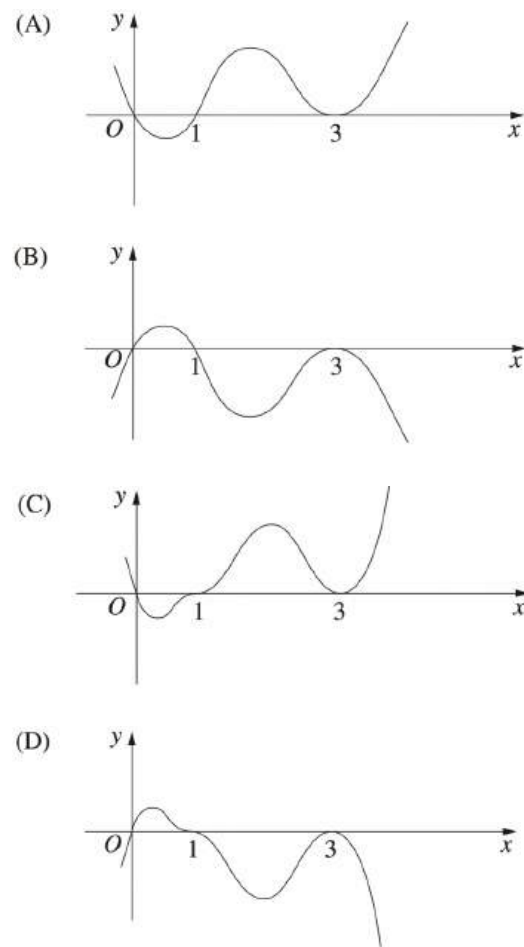
A monic polynomial $p(x)$ of degree 4 has one repeated zero of multiplicity 2 and is divisible by $x^2 + x + 1$.

Which of the following could be the graph of $p(x)$?



9. Functions, EXT1 F2 2013 HSC 4 MC

Which diagram best represents the graph $y = x(1-x)^3(3-x)^2$?



10. Functions, EXT1 F2 2014 HSC 5 MC

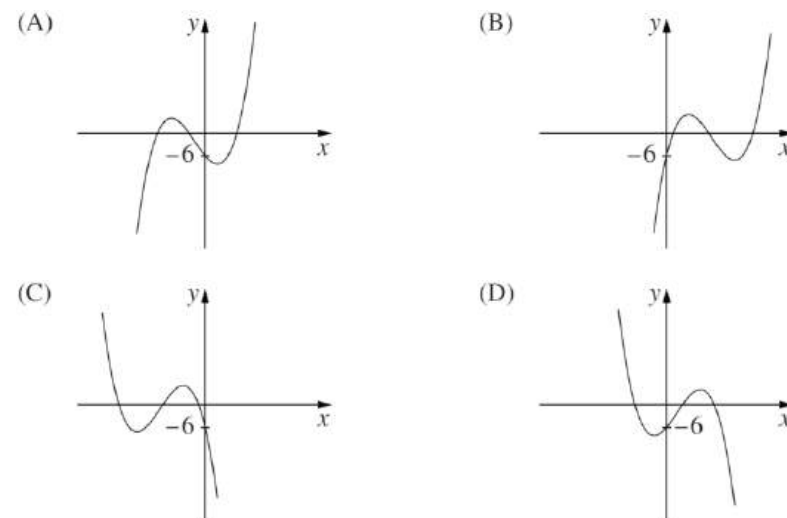
Which group of three numbers could be the roots of the polynomial equation $x^3 + ax^2 - 41x + 42 = 0$?

- A. 2, 3, 7
- B. 1, -6, 7
- C. -1, -2, 21
- D. -1, -3, -14

11. Functions, EXT1 F2 2016 HSC 10 MC

Consider the polynomial $p(x) = ax^3 + bx^2 + cx - 6$ with a and b positive.

Which graph could represent $p(x)$?



12. Functions, EXT1 F2 2019 HSC 7 MC

Let $P(x) = qx^3 + rx^2 + rx + q$ where q and r are constants, $q \neq 0$. One of the zeros of $P(x)$ is -1 .

Given that α is a zero of $P(x)$, $\alpha \neq -1$, which of the following is also a zero?

- A. $-\frac{1}{\alpha}$
- B. $-\frac{q}{\alpha}$
- C. $\frac{1}{\alpha}$
- D. $\frac{q}{\alpha}$

13. Functions, EXT1' F2 2016 HSC 2 MC

Which polynomial has a multiple root at $x = 1$?

- A. $x^5 - x^4 - x^2 + 1$
- B. $x^5 - x^4 - x - 1$
- C. $x^5 - x^3 - x^2 + 1$
- D. $x^5 - x^3 - x + 1$

14. Functions, EXT1 F2 2013 HSC 11a

The polynomial equation $2x^3 - 3x^2 - 11x + 7 = 0$ has roots α , β and γ .

Find $\alpha\beta\gamma$. (1 mark)

15. Functions, EXT1' F2 2014 HSC 14a

Let $P(x) = x^5 - 10x^2 + 15x - 6$.

Show that $x = 1$ is a root of $P(x)$ of multiplicity three. (2 marks)

16. Functions, EXT1' F2 2017 HSC 13b

Let a , b and c be real numbers. Suppose that $P(x) = x^4 + ax^3 + bx^2 + cx + 1$ has roots α , $\frac{1}{\alpha}$, β , $\frac{1}{\beta}$, where $\alpha > 0$ and $\beta > 0$.

Prove that $a = c$. (2 marks)

17. Functions, EXT1' F2 2018 HSC 11b

The polynomial $p(x) = x^3 + ax^2 + b$ has a zero at r and a double zero at 4.

Find the values of a , b and r . (3 marks)

18. Functions, EXT1 F2 2008 HSC 2c

The polynomial $p(x)$ is given by $p(x) = ax^3 + 16x^2 + cx - 120$, where a and c are constants.

The three zeros of $p(x)$ are -2 , 3 and β .

Find the value of β . (3 marks)

19. Functions, EXT1' F2 2017 HSC 12d

Let $P(x)$ be a polynomial.

i. Given that $(x - \alpha)^2$ is a factor of $P(x)$, show that

$$P(\alpha) = P'(\alpha) = 0. \text{ (2 marks)}$$

ii. Given that the polynomial $P(x) = x^4 - 3x^3 + x^2 + 4$ has a factor $(x - \alpha)^2$, find the value of α . (2 marks)

20. Functions, EXT1 F2 SM-Bank 2

The polynomial $P(x) = x^3 - 2x^2 + kx + 24$ has roots α , β , γ .

i. Find the value of $\alpha + \beta + \gamma$. (1 mark)

ii. Find the value of $\alpha\beta\gamma$. (1 mark)

iii. It is known that two of the roots are equal in magnitude but opposite in sign.

Find the third root and hence find the value of k . (2 marks)

21. Functions, EXT1' F2 2013 HSC 15b

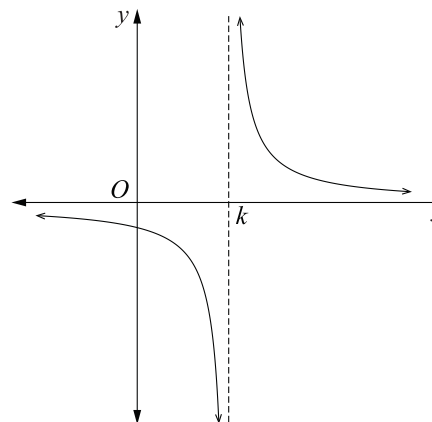
The polynomial $P(x) = ax^4 + bx^3 + cx^2 + e$ has remainder -3 when divided by $x - 1$. The polynomial has a double root at $x = -1$.

i. Show that $4a + 2c = -\frac{9}{2}$. (2 marks)

ii. Hence, or otherwise, find the slope of the tangent to the graph $y = P(x)$ when $x = 1$. (1 mark)

22. Functions, EXT1 F2 2019 HSC 14b

The diagram shows the graph of $y = \frac{1}{x - k}$, where k is a positive real number.



By considering the graphs of $y = x^2$ and $y = \frac{1}{x - k}$, explain why the function $f(x) = x^3 - kx^2 - 1$ has exactly one real zero. (2 marks)

23. Functions, EXT1 F2 2021 HSC 11h

The roots of $x^4 - 3x + 6 = 0$ are α , β , γ and δ .

What is the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$? (2 marks)

24. Functions, EXT1 F2 EQ-Bank 1

The polynomial $P(x) = 2x^3 - mx^2 + nx + 27$ has a double root and $P(-3) = P'(-3) = 0$.

Find the values of m and n and hence find the other root of $P(x)$. (3 marks)

25. Functions, EXT1 F2 EQ-Bank 2

In the equation $x^3 - 8x^2 + 11x + 20 = 0$ one of the roots is equal to the sum of the other two roots.

Find the value of the three roots. **(3 marks)**

26. Functions, EXT1 F2 SM-Bank 1

The equation $2x^3 + x^2 - kx + 6 = 0$ has 2 roots which are reciprocals of each other.

Find the value of k . **(2 marks)**

27. Functions, EXT1' F2 2007 HSC 3b

The zeros of $x^3 - 5x + 3$ are α, β and γ .

Find a cubic polynomial with integer coefficients whose zeros are $2\alpha, 2\beta$ and 2γ . **(2 marks)**

28. Functions, EXT1' F2 2009 HSC 3c

Let $P(x) = x^3 + ax^2 + bx + 5$, where a and b are real numbers.

Find the values of a and b given that $(x - 1)^2$ is a factor of $P(x)$. **(3 marks)**

29. Functions, EXT1 F2 SM-Bank 5

The polynomial $P(x) = x^3 + px^2 + qx + r$ has roots $\sqrt{k}, -\sqrt{k}$ and α .

- Explain why $\alpha + p = 0$. **(1 mark)**
 - Show that $k\alpha = r$. **(1 mark)**
 - Show that $pq = r$. **(2 marks)**
-

30. Functions, EXT1 F2 2006 HSC 4a

The cubic polynomial $P(x) = x^3 + rx^2 + sx + t$, where r, s and t are real numbers, has three real zeros, $1, \alpha$ and $-\alpha$.

- Find the value of r . **(1 mark)**
 - Find the value of $s + t$. **(2 marks)**
-

31. Functions, EXT1' F2 2016 HSC 13d

Suppose $p(x) = ax^3 + bx^2 + cx + d$ with a, b, c and d real, $a \neq 0$.

- Deduce that if $b^2 - 3ac < 0$ then $p(x)$ cuts the x -axis only once. **(2 marks)**
 - If $b^2 - 3ac = 0$ and $p\left(-\frac{b}{3a}\right) = 0$, what is the multiplicity of the root $x = -\frac{b}{3a}$? **(2 marks)**
-

32. Functions, EXT1' F2 2019 HSC 16ai

Consider the equation $x^3 - px + q = 0$, where p and q are real numbers and $p > 0$.

Let $r = \sqrt{\frac{4p}{3}}$ and $\cos 3\theta = \frac{-4q}{r^3}$.

Show that $r\cos\theta$ is a root of $x^3 - px + q = 0$.

You may use the result $4\cos^3\theta - 3\cos\theta = \cos 3\theta$. (Do NOT prove this.) **(2 marks)**

33. Functions, EXT1 F2 2022 HSC 13d

The monic polynomial, P , has degree 3 and roots α, β, γ .

It is given that

$$\alpha^2 + \beta^2 + \gamma^2 = 85 \quad \text{and}$$

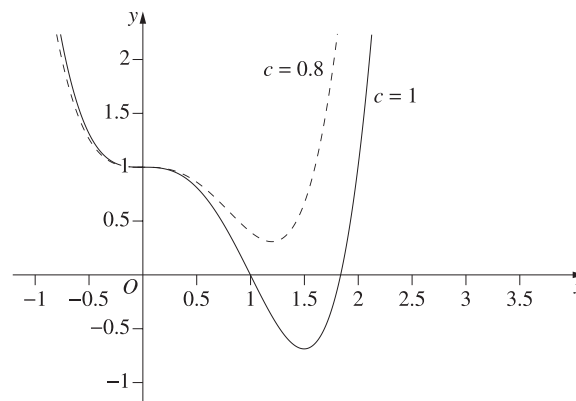
$$P'(\alpha) + P'(\beta) + P'(\gamma) = 87.$$

Find $\alpha\beta + \beta\gamma + \gamma\alpha$. **(3 marks)**

34. Functions, EXT1 F2 2023 HSC 14b

Consider the hyperbola $y = \frac{1}{x}$ and the circle $(x - c)^2 + y^2 = c^2$, where c is a constant.

- Show that the x -coordinates of any points of intersection of the hyperbola and circle are zeros of the polynomial $P(x) = x^4 - 2cx^3 + 1$. **(1 mark)**
- The graphs of $y = x^4 - 2cx^3 + 1$ for $c = 0.8$ and $c = 1$ are shown.



By considering the given graphs, or otherwise, find the exact value of $c > 0$ such that the hyperbola $y = \frac{1}{x}$ and the circle $(x - c)^2 + y^2 = c^2$ intersect at only one point. **(3 marks)**

35. Trigonometry, EXT1 T3 2020 HSC 14b

i. Show that $\sin^3\theta - \frac{3}{4}\sin\theta + \frac{\sin(3\theta)}{4} = 0$. (2 marks)

ii. By letting $x = 4\sin\theta$ in the cubic equation $x^3 - 12x + 8 = 0$.

Show that $\sin(3\theta) = \frac{1}{2}$. (2 marks)

iii. Prove that $\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} = \frac{3}{2}$. (3 marks)

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Worked Solutions

1. Functions, EXT1' F2 2015 HSC 4 MC

$$\begin{aligned} P(x) &= 3x^2 + 2x - 5 \\ &= (3x + 5)(x - 1) \end{aligned}$$

$\therefore$ Only possible double roots occur when

$$x = 1 \quad \text{or} \quad x = -\frac{5}{3}$$

$$P(1) = 1 + 1 - 5 + 3 = 0$$

$\therefore x = 1$ is double root.

$\Rightarrow C$

2. Functions, EXT1 F2 2024 HSC 1 MC

$$\alpha\beta\gamma = -\frac{d}{a} = 6$$

$$-1 \times -3 \times \alpha = 6$$

$$\alpha = 2$$

$\Rightarrow B$

3. Functions, EXT1 F2 2012 HSC 3 MC

Using $ax^3 + bx^2 + cx + d = 0$

By elimination

$$\alpha\beta\gamma = -\frac{d}{a} = 1$$

$\therefore$ Cannot be A or C (where $\alpha\beta\gamma = -1$)

$$\alpha + \beta + \gamma = -\frac{b}{a} = -2$$

$\therefore$ Cannot be D (where $\alpha + \beta + \gamma = 2$)

$\Rightarrow B$

4. Functions, EXT1' F2 2012 HSC 5 MC

$$\alpha\beta\gamma = \frac{-d}{a} = \frac{1}{2}$$

$$\therefore \alpha^3\beta^3\gamma^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$\therefore \frac{1}{\alpha^3\beta^3\gamma^3} = 8$$

$\Rightarrow C$

5. Functions, EXT1' F2 2017 HSC 5 MC

Since α, β, γ are roots,

$p(\alpha) = \alpha^3 - 2\alpha + 2 = 0 \quad \dots (1)$

$p(\beta) = \beta^3 - 2\beta + 2 = 0 \quad \dots (2)$

$p(\gamma) = \gamma^3 - 2\gamma + 2 = 0 \quad \dots (3)$

Add: $(1) + (2) + (3)$

$\alpha^3 + \beta^3 + \gamma^3 - 2(\alpha + \beta + \gamma) + 6 = 0$

$\therefore \alpha^3 + \beta^3 + \gamma^3 = -6 \quad (\text{note: } \alpha + \beta + \gamma = -\frac{b}{a} = 0)$

$\Rightarrow B$

6. Functions, EXT1' F2 2018 HSC 3 MC

$\alpha\beta\gamma = 1$

$\alpha\beta + \beta\gamma + \gamma\alpha = 5$

$\alpha + \beta + \gamma = -2$

If roots are $\frac{-1}{\alpha}, \frac{-1}{\beta}, \frac{-1}{\gamma}$:

$\frac{-1}{\alpha} \cdot \frac{-1}{\beta} \cdot \frac{-1}{\gamma} = \frac{-1}{\alpha\beta\gamma} = -1$

$\Rightarrow d = 1$

$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{(\alpha + \beta + \gamma)}{\alpha\beta\gamma} = -2$

$\Rightarrow c = -2$

$\frac{-1}{\alpha} - \frac{-1}{\beta} - \frac{-1}{\gamma} = \frac{-(\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma} = -5$

$\Rightarrow b = 5$

$\therefore$ Equation is $x^3 + 5x^2 - 2x + 1 = 0$

$\Rightarrow D$

7. Functions, EXT1' F2 2019 HSC 4 MC

$f(x) = 2x^3 + bx^2 + cx + d$

$f'(x) = 6x^2 + 2bx + c$

Roots at 1 and -3 :

$f(1) = 0$

$2 + b + c + d = 0$

$b + c + d = -2 \quad \dots (1)$

$f(-3) = 0$

$-54 + 9b - 3c + d = 0$

$9b - 3c + d = 54 \quad \dots (2)$

$(2) - (1)$

$8b - 4c = 56$

$2b - c = 14 \quad \dots (3)$

If double root at 1:

$f'(1) = 0$

$6 + 2b + c = 0$

$2b + c = -6 \quad \dots (4)$

$(3) + (4)$

$4b = 8$

$b = 2$

If double root at -3 :

$f'(-3) = 0$

$54 - 6b + c = 0$

$-6b + c = -54 \quad \dots (5)$

$(3) + (5)$

$-4b = -40$

$b = 10$

$\Rightarrow D$

8. Functions, EXT1 F2 2020 HSC 5 MC

Since $p(x)$ is monic,
$$\Rightarrow p(x) = (x-a)^2(x^2+x+1)$$

Consider x^2+x+1
$$\Delta = \sqrt{1^2-4 \cdot 1 \cdot 1} = \sqrt{-3} < 0 \Rightarrow \text{No roots}$$

 $\therefore$ Only root is $x = a$ (multiplicity 2)
 $\Rightarrow$ Eliminate B and D
As $x \rightarrow \infty, p(x) \rightarrow \infty$
 $\Rightarrow C$

9. Functions, EXT1 F2 2013 HSC 4 MC

$$y = x(1-x)^3(3-x)^2$$

By elimination
Consider when $x < 0$,
$$y = (-\text{ve}) \times (+\text{ve}) \times (+\text{ve}) < 0$$

 $\therefore$ Cannot be A or C
Consider the cubic factor $(1-x)^3$,
The graph must have a stationary point at $x = 1$
 $\therefore$ Cannot be B
 $\Rightarrow D$

10. Functions, EXT1 F2 2014 HSC 5 MC

Let roots be α, β, γ
$$\alpha\beta\gamma = -\frac{d}{a} = -42$$

 $\therefore$ Cannot be A or C
$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = -41$$

 $\therefore$ Cannot be D
 $\Rightarrow B$

11. Functions, EXT1 F2 2016 HSC 10 MC

$$p(x) = ax^3 + bx^2 + cx - 6$$
$$p'(x) = 3ax^2 + 2bx + c$$
$$p''(x) = 6ax + 2b$$

Since $a, b > 0$,
As $x \rightarrow \infty, p(x) \rightarrow \infty$
 $\therefore$ Eliminate C and D .

POI occurs when $p''(x) = 0$,
$$6ax + 2b = 0$$
$$x = -\frac{b}{3a} < 0, \quad (a, b > 0)$$

 $\therefore$ Eliminate B
 $\Rightarrow A$

12. Functions, EXT1 F2 2019 HSC 7 MC

Roots: $\alpha, \beta, -1$
$$\alpha\beta(-1) = -\frac{d}{a} = -\frac{q}{q} = -1$$

$$-\alpha\beta = -1$$
$$\therefore \beta = \frac{1}{\alpha}$$

 $\Rightarrow C$

Mean mark 51%.

13. Functions, EXT1' F2 2016 HSC 2 MC

Consider C ,
$$P(1) = 1-1-1+1 = 0$$
$$P'(x) = 5x^4-3x^2-2x$$
$$P'(1) = 5-3-2 = 0$$

 $\therefore$ Multiple root at $x = 1$
 $\Rightarrow C$

14. Functions, EXT1 F2 2013 HSC 11a

$$P(x) = 2x^3 - 3x^2 - 11x + 7 = 0$$
$$\alpha\beta\gamma = -\frac{d}{a} = -\frac{7}{2}$$

15. Functions, EXT1' F2 2014 HSC 14a

$$P(x) = x^5 - 10x^2 + 15x - 6$$
$$P(1) = 1 - 10 + 15 - 6 = 0$$

$$P'(x) = 5x^4 - 20x + 15$$
$$P'(1) = 5 - 20 + 15 = 0$$
$$P''(x) = 20x^3 - 20$$
$$P''(1) = 20 - 20 = 0$$
$$P'''(x) = 60x^2$$
$$P'''(1) = 60 \neq 0$$

$\therefore x = 1$ is a root of $P(x)$, of multiplicity 3.

16. Functions, EXT1' F2 2017 HSC 13b

$$\text{Sum of roots} = -\frac{b}{a}:$$
$$-a = \alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta}$$

$$\text{Sum of products of 3 roots} = -\frac{d}{a}:$$
$$-c = \alpha \cdot \frac{1}{\alpha} \cdot \beta + \alpha \cdot \frac{1}{\alpha} \cdot \frac{1}{\beta} + \alpha \cdot \beta \cdot \frac{1}{\beta} + \frac{1}{\alpha} \cdot \beta \cdot \frac{1}{\beta}$$
$$= \beta + \frac{1}{\beta} + \alpha + \frac{1}{\alpha}$$
$$= -a$$

$\therefore a = c \dots$ as required.

17. Functions, EXT1' F2 2018 HSC 11b

$$p(x) = x^3 + ax^2 + b$$
$$p'(x) = 3x^2 + 2ax$$

Double root at 4:

$$p'(4) = 0$$
$$3 \times 16 + 8a = 0$$
$$a = -6$$

$$p(4) = 0$$
$$64 + 16a + b = 0$$
$$64 - 96 + b = 0$$
$$b = 32$$

Roots of $p(x)$ are $4, 4, r$

$$4 + 4 + r = -\frac{a}{1} = 6$$
$$\therefore r = -2$$

18. Functions, EXT1 F2 2008 HSC 2c

$p(x) = ax^3 + 16x^2 + cx - 120$

Roots: $-2, 3, \beta$

$$-2 + 3 + \beta = -\frac{B}{A}$$

$$\beta + 1 = -\frac{16}{a}$$

$$\beta = -\frac{16}{a} - 1 \quad \dots (1)$$

$$-2 \times 3 \times \beta = -\frac{D}{A}$$

$$-6\beta = \frac{120}{a}$$

$$\beta = -\frac{20}{a} \quad \dots (2)$$

$$-\frac{16}{a} - 1 = -\frac{20}{\beta}$$

$$\frac{-16-a}{a} = \frac{-20}{\beta}$$

Substitute $a = 4$ into (1)

$$\begin{aligned} \therefore \beta &= -\frac{16}{4} - 1 \\ &= -5 \end{aligned}$$

MARKER'S COMMENT: Many students displayed significant inefficiencies in solving simultaneous equations.

19. Functions, EXT1' F2 2017 HSC 12d

i. $P(x) = (x-\alpha)^2 \cdot Q(x)$

$$\begin{aligned} P'(x) &= 2(x-\alpha) \cdot Q(x) + (x-\alpha)^2 \cdot Q'(x) \\ &= (x-\alpha)[2Q(x) + (x-\alpha) \cdot Q'(x)] \end{aligned}$$

$$P(\alpha) = 0 \times Q(\alpha) = 0$$

$$P'(\alpha) = 0[2Q(\alpha) + 0 \times Q'(\alpha)] = 0$$

$$\therefore P(\alpha) = P'(\alpha) = 0 \dots \text{as required.}$$

ii. $P(x) = x^4 - 3x^3 + x^2 + 4$

$$\begin{aligned} P'(x) &= 4x^3 - 9x^2 + 2x \\ &= x(4x^2 - 9x + 2) \\ &= x(4x-1)(x-2) \end{aligned}$$

$$\therefore P'(x) = 0 \text{ when } x = 0, \frac{1}{4} \text{ or } 2$$

$$\Rightarrow \text{Multiple roots may exist at } x = 0, \frac{1}{4} \text{ or } 2.$$

Test each root in $P(x)$:

$$P(0) = 0 - 0 + 0 + 4 = 4$$

$$P\left(\frac{1}{4}\right) = \left(\frac{1}{4}\right)^4 - 3\left(\frac{1}{4}\right)^3 + \left(\frac{1}{4}\right)^2 + 4 = 4\frac{5}{256}$$

$$P(2) = 16 - 3(8) + 4 + 4 = 0$$

$$\therefore (x-2)^2 \text{ is a factor of } P(x)$$

$$\therefore \alpha = 2$$

20. Functions, EXT1 F2 SM-Bank 2

i. $\alpha + \beta + \gamma = -\frac{b}{a} = 2$

ii. $\alpha\beta\gamma = -\frac{d}{a} = -24$

iii. Let roots be $\alpha, -\alpha, \beta$:

$$\begin{aligned}\alpha - \alpha + \beta &= 2 \\ \beta &= 2\end{aligned}$$

Substitute $\beta = 2$ into equation:

$$\begin{aligned}2^3 - 2 \cdot 2^2 + 2k + 24 &= 0 \\ 2k &= -24 \\ k &= -12\end{aligned}$$

21. Functions, EXT1' F2 2013 HSC 15b

i. $P(x) = ax^4 + bx^3 + cx^2 + e$
 $P'(x) = 4ax^3 + 3bx^2 + 2cx$

$$\begin{aligned}P(1) &= -3 \\ a + b + c + e &= -3 \quad \dots (1) \\ P(-1) &= 0 \\ a - b + c + e &= 0 \quad \dots (2) \\ P'(-1) &= 0 \\ -4a + 3b - 2c &= 0 \quad \dots (3)\end{aligned}$$

$$\begin{aligned}(1) - (2) \\ 2b &= -3 \\ b &= -\frac{3}{2}\end{aligned}$$

Substitute into (3)

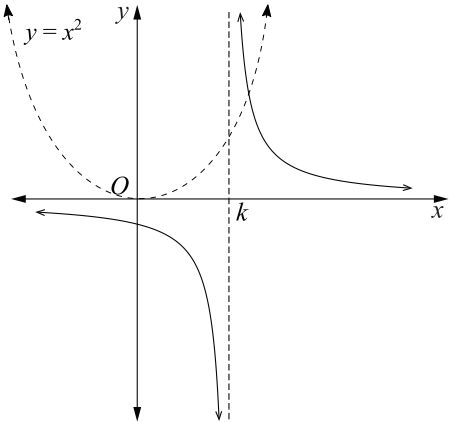
$$\begin{aligned}\therefore 4a + 2c &= 3b \\ &= -\frac{9}{2} \quad \dots \text{as required}\end{aligned}$$

ii. $P'(1) = 4a + 3b + 2c$

$$\begin{aligned}&= -\frac{9}{2} - \frac{9}{2} \\ &= -9\end{aligned}$$

22. Functions, EXT1 F2 2019 HSC 14b

Draw $y = x^2$ on the diagram:



One point of intersection occurs when

$$x^2 = \frac{1}{x-k}$$
$$x^3 - kx^2 = 1$$
$$x^3 - kx^2 - 1 = 0$$

Since only 1 point of intersection
 $\Rightarrow x^3 - kx^2 - 1 = 0$ has exactly 1 zero

23. Functions, EXT1 F2 2021 HSC 11h

$$\alpha\beta\gamma\delta = \frac{6}{1} = 6$$
$$\alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \delta\alpha\beta = -\frac{(-3)}{1} = 3$$
$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{\beta\gamma\delta + \alpha\gamma\delta + \beta\gamma\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta}$$
$$= \frac{3}{6}$$
$$= \frac{1}{2}$$

24. Functions, EXT1 F2 EQ-Bank 1

$$P(x) = 2x^3 - mx^2 + nx + 27$$
$$P'(x) = 6x^2 - 2mx + n$$
$$P(-3) = 2 \times (-3)^3 - (-3)^2m - 3n + 27$$
$$0 = -54 - 9m - 3n + 27$$
$$9m + 3n = -27 \dots (1)$$

$$P'(-3) = 6 \times (-3)^2 + 6m + n$$
$$6m + n = -54$$
$$18m + 3n = -162 \dots (2)$$

(2) - (1):

$$9m = -135 \Rightarrow m = -15$$
$$6 \times -15 + n = -54 \Rightarrow n = 36$$

Using product of roots:

$$\alpha\beta\gamma = \frac{-d}{a}$$
$$(-3)^2\gamma = \frac{-27}{2}$$
$$\gamma = \frac{-27}{2 \times 9} = -\frac{3}{2}$$

25. Functions, EXT1 F2 EQ-Bank 2

$$x^3 - 8x^2 + 11x + 20 = 0$$

Sum of roots $= -\frac{b}{a}:$

$$\alpha + \beta + (\alpha + \beta) = 8$$

$$2\alpha + 2\beta = 8$$

$$\beta = 4 - \alpha \dots (1)$$

Product of roots $= -\frac{d}{a}:$

$$\alpha\beta(\alpha + \beta) = -20 \dots (2)$$

Substitute (1) into (2):

$$\alpha(4 - \alpha)(4) = -20$$

$$16\alpha - 4\alpha^2 = -20$$

$$\alpha^2 - 4\alpha - 5 = 0$$

$$(\alpha - 5)(\alpha + 1) = 0$$

$$\alpha = 5 \text{ or } -1$$

$$\beta = 5 \text{ or } -1 \text{ (using (1))}$$

$$\alpha + \beta = 5 - 1 = 4$$

$$\therefore \text{Roots} = -1, 4, 5$$

26. Functions, EXT1 F2 SM-Bank 1

Let roots be $\alpha, \frac{1}{\alpha}, \beta:$

$$\alpha \cdot \frac{1}{\alpha} \cdot \beta = -\frac{d}{a}$$

$$\beta = -\frac{6}{2}$$

$$= -3$$

Substitute $\beta = -3$ into equation:

$$2(-3)^3 + (-3)^2 - (-3)k + 6 = 0$$

$$-54 + 9 + 3k + 6 = 0$$

$$3k = 39$$

$$k = 13$$

COMMENT: Substituting $\beta = -3$ into the equation solves this question efficiently.

27. Functions, EXT1' F2 2007 HSC 3b

Solution 1

$$\alpha + \beta + \gamma = 0$$

$$\therefore 2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma) \\ = 0$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = -5$$

$$\therefore 2\alpha 2\beta + 2\beta 2\gamma + 2\gamma 2\alpha = 4(\alpha\beta + \beta\gamma + \gamma\alpha) \\ = -20$$

$$\alpha\beta\gamma = -3$$

$$\therefore 2\alpha 2\beta 2\gamma = 8\alpha\beta\gamma \\ = -24$$

$$\therefore \text{Polynomial is } x^3 - 20x + 24$$

Solution 2

$$P(x) = x^3 - 5x + 3$$

New zeros are $2\alpha, 2\beta$ and 2γ

$$\therefore H(x) = \left(\frac{x}{2}\right)^3 - 5\left(\frac{x}{2}\right) + 3 \\ = \frac{x^3}{8} - \frac{5x}{2} + 3$$

$$\therefore \text{New Polynomial with integer coefficients is } \\ x^3 - 20x + 24$$

28. Functions, EXT1' F2 2009 HSC 3c

$$P(x) = x^3 + ax^2 + bx + 5$$

$$P(1) = 1 + a + b + 5 = 0$$

$$\therefore b = -a - 6 \dots (1)$$

$$P'(x) = 3x^2 + 2ax + b$$

$$P'(1) = 3 + 2a + b = 0$$

$$2a + b = -3 \dots (2)$$

Substitute $b = -a - 6$ from (1) into (2)

$$2a + (-a - 6) = -3$$

$$\therefore a = 3, \quad b = -9$$

29. Functions, EXT1 F2 SM-Bank 5

i. Sum of roots:

$$\begin{aligned}\sqrt{k}-\sqrt{k}+\alpha &= -\frac{b}{a} \\ \alpha &= -p \\ \alpha + p &= 0\end{aligned}$$

ii. Product of roots:

$$\begin{aligned}\sqrt{k}\times -\sqrt{k} \times \alpha &= -\frac{d}{a} \\ -k\alpha &= -r \\ \therefore k\alpha &= r\end{aligned}$$

iii. $\sqrt{k}\Big(-\sqrt{k}\Big)+\sqrt{k}\alpha-\sqrt{k}\alpha=\frac{c}{d}$

$$-k=q$$

Substitute $k=-q$ into part (ii):

$$\begin{aligned}-q\alpha &= r \\ -q\times -p &= r \quad (\text{using part(i)}) \\ \therefore pq &= r\end{aligned}$$

30. Functions, EXT1 F2 2006 HSC 4a

i. $P(x)=x^3+rx^2+sx+t$

Roots are $1,\alpha,-\alpha$

$$\begin{aligned}1+\alpha+-\alpha &= -\frac{b}{a} \\ 1 &= -\frac{r}{1} \\ \therefore r &= -1\end{aligned}$$

ii. $P(x)=x^3-x^2+sx+t$

$$P(1)=0$$

$$0=1-1+(s\times 1)+t$$

$$0=s+t$$

$$\therefore \text{ Value of } s+t=0$$

31. Functions, EXT1' F2 2016 HSC 13d

i. $p(x)=ax^3+bx^2+cx+d$

$$p'(x)=3ax^2+2bx+c$$

$\Rightarrow p(x)$ will cut the x -axis once

$$\text{only if } \Delta(p'(x))<0$$

$$(2b)^2-4(3a)c<0$$

$$4b^2-12ac<0$$

$$b^2-3ac<0$$

ii. $p\Big(-\frac{b}{3a}\Big)=0$

$$\begin{aligned}p'\Big(-\frac{b}{3a}\Big) &= 3a\Big(-\frac{b}{3a}\Big)^2+2b\Big(-\frac{b}{3a}\Big)+c \\ &= -\frac{b^2}{3a}+c \\ &= 0 \quad (\text{given } b^2-3ac=0)\end{aligned}$$

$\therefore$ Multiplicity at least 2.

$$p''(x)=6ax+2b$$

$$p''\Big(-\frac{b}{3a}\Big)=6a\Big(-\frac{b}{3a}\Big)+2b=0$$

$\therefore$ Multiplicity of $x=-\frac{b}{3a}$ is 3.

♦♦ Mean mark 32%.

32. Functions, EXT1' F2 2019 HSC 16ai

$$x^3-px+q=0 \dots (1)$$

$$r=\sqrt{\frac{4p}{3}} \Rightarrow p=\frac{3r^2}{4}$$

$$\cos 3\theta=\frac{-4q}{r^3} \Rightarrow q=\frac{-r^3\cos 3\theta}{4}$$

$$\text{Given } 4\cos^3\theta-3\cos\theta=\cos 3\theta \dots (2)$$

Substitute $x=r\cos\theta$ into (1)

$$r^3\cos^3\theta-\frac{3r^2}{4}r\cos\theta-\frac{r^3\cos 3\theta}{4}=0$$

$$\frac{r^3}{4}\underbrace{(4\cos^3\theta-3\cos\theta-\cos 3\theta)}_{=0 \text{ (see (2) above)}}=0$$

$\therefore r\cos\theta$ is a root.

♦ Mean mark 49%.

33. Functions, EXT1 F2 2022 HSC 13d

$P(x) = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma$

$P'(x) = 3x^2 - 2(\alpha + \beta + \gamma)x + \alpha\beta + \alpha\gamma + \beta\gamma$

$P'(\alpha) + P'(\beta) + P'(\gamma)$

$$\begin{aligned} &= 3\alpha^2 - 2(\alpha + \beta + \gamma)\alpha + \alpha\beta + \alpha\gamma + \beta\gamma \\ &\quad + 3\beta^2 - 2(\alpha + \beta + \gamma)\beta + \alpha\beta + \alpha\gamma + \beta\gamma \\ &\quad + 3\gamma^2 - 2(\alpha + \beta + \gamma)\gamma + \alpha\beta + \alpha\gamma + \beta\gamma \\ &= 3(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha^2 + \beta^2 + \gamma^2) \end{aligned}$$

$$\begin{aligned} &- 2(\alpha\beta + \alpha\gamma + \alpha\beta + \beta\gamma + \alpha\gamma + \beta\gamma) + 3(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= (\alpha^2 + \beta^2 + \gamma^2) - (\alpha\beta + \alpha\gamma + \beta\gamma) \end{aligned}$$

Substituting in given values:

$$\begin{aligned} 85 - (\alpha\beta + \alpha\gamma + \beta\gamma) &= 87 \\ \therefore \alpha\beta + \alpha\gamma + \beta\gamma &= -2 \end{aligned}$$

♦♦ Mean mark 33%.

34. Functions, EXT1 F2 2023 HSC 14b

i. $y = \frac{1}{x} \dots (1)$

$(x - c)^2 + y^2 = c^2 \dots (2)$

Substitute (1) into (2):

$(x - c)^2 + \left(\frac{1}{x}\right)^2 = c^2$

$x^2 - 2cx + c^2 + \frac{1}{x^2} = c^2$

$x^4 - 2cx^3 + 1 = 0$

- ii. The two graphs show that for some value of $0.8 \leq c \leq 1$, $P(x)$ has a minimum that touches the x -axis once.

$P(x) = x^4 - 2cx^3 + 1$

$P'(x) = 4x^3 - 6cx^2$

Find x when $P'(x) = 0$:

$4x^3 - 6cx^2 = 0$

$2x^2(2x - 3c) = 0$

$x = \frac{3c}{2} \quad (x \neq 0)$

Find c when $P(\frac{3c}{2}) = 0$:

$\left(\frac{3c}{2}\right)^4 - 2c\left(\frac{3c}{2}\right)^3 + 1 = 0$

$\frac{81c^4}{16} - \frac{54c^4}{8} + 1 = 0$

$\frac{(108 - 81)c^4}{16} = 1$

$\frac{27c^4}{16} = 1$

$c^4 = \frac{16}{27}$

$c = \sqrt[4]{\frac{16}{27}}$

≈ 0.877

Mean mark (i) 53%.

Mean mark (ii) 19%.

35. Trigonometry, EXT1 T3 2020 HSC 14b

i. Prove: $\sin^3\theta - \frac{3}{4}\sin\theta + \frac{\sin(3\theta)}{4} = 0$

$$\begin{aligned} \text{LHS} &= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}(\sin2\theta\cos\theta + \cos2\theta\sin\theta) \\ &= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}(2\sin\theta\cos^2\theta + \sin\theta(1-2\sin^2\theta)) \\ &= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}(2\sin\theta(1-\sin^2\theta) + \sin\theta-2\sin^3\theta) \\ &= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{1}{4}(2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta) \\ &= \sin^3\theta - \frac{3}{4}\sin\theta + \frac{3}{4}\sin\theta - \sin^3\theta \\ &= 0 \end{aligned}$$

ii. Show $\sin(3\theta) = \frac{1}{2}$

Using part (i):

$$\begin{aligned} \frac{\sin(3\theta)}{4} &= \frac{3}{4}\sin\theta - \sin^3\theta \\ \sin(3\theta) &= 3\sin\theta - 4\sin^3\theta \dots (1) \end{aligned}$$

$$x^3 - 12x + 8 = 0$$

Let $x = 4\sin\theta$

$$\begin{aligned} (4\sin\theta)^3 - 12(4\sin\theta) + 8 &= 0 \\ 64\sin^3\theta - 48\sin\theta &= 0 \\ -16(\underbrace{3\sin\theta - 4\sin^2\theta}_{\text{see (1) above}}) &= -8 \\ -16\sin(3\theta) &= -8 \\ \sin(3\theta) &= \frac{1}{2} \end{aligned}$$

iii. Prove: $\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} = \frac{3}{2}$

Solutions to $x^3 - 12x + 8 = 0$ are

$$x = 4\sin\theta \text{ where } \sin(3\theta) = \frac{1}{2}$$

When $\sin3\theta = \frac{1}{2}$,

$$3\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}, \dots$$

$$\theta = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}, \dots$$

$\therefore$ Solutions

$$x = 4\sin \frac{\pi}{18} \quad \left(= 4\sin \frac{17\pi}{18} \right)$$

$$x = 4\sin \frac{5\pi}{18} \quad \left(= 4\sin \frac{13\pi}{18} \right)$$

$$x = 4\sin \frac{25\pi}{18} \quad \left(= 4\sin \frac{29\pi}{18} \right)$$

If roots of $x^3 - 12x + 8 = 0$ are α, β, γ :

$$\alpha + \beta + \gamma = -\frac{b}{a} = 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = -12$$

$$\left(4\sin \frac{\pi}{18}\right)^2 + \left(4\sin \frac{5\pi}{18}\right)^2 + \left(4\sin \frac{25\pi}{18}\right)^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$16\left(\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18}\right) = 0 - 2(-12)$$

$$\therefore \sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} = \frac{24}{16} = \frac{3}{2}$$